

Virtual Interview Simulator Using AI for Enhancing Technical Communication Skills

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Abstract— Our pioneering initiative introduces the first open-source module for job interview training within a virtual metaverse, leveraging NLP-based Generative AI, designed for extension and testing by the scientific community, thereby enhancing insights into the limitations and ethical considerations of AI-driven interview simulation platforms.

Keywords— Metaverse, Generative AI, Interview stimulation, Large Language models, Natural Language Processing.

Highlights

- *Learners invest less time in revision of AI-generated texts.*
- *Trust in AI-based text generators leads to shorter duration of revision.*
- *Frequent perception of media reports about AI leads to shorter duration of revision.*
- *Learners with more experience in academic writing revise more thoroughly.*
- *Female learners tend to revise more thoroughly than male learners.*

I. INTRODUCTION

A trend among today's youth is the rise of NEETs (Not in Education Employment or training), they have low self-confidence, poor socio-emotional skills, and weak interaction skills. This affects their performance in interviews as interviewers assess not only the practical application of skills but also the clarity and confidence with which they are presented.

To address this issue, many countries have adapted a training center with professional personnel to secure jobs for the youth. The issue with this is that it is time consuming and costly. To reduce these major factors, we present a job interview simulation environment which uses a social virtual character as a recruiter and signal processing techniques to enable the virtual character to react and adapt to the user's behavior and emotions.

In section II we discuss the literature review the following section III shows the methods of this paper. Section IV shows the results and discussions we made followed by the last section V which is the conclusion and recommendations.

The purpose of this simulation is to help youngsters improve social skills which are pertinent to job interviews. The system we propose features a real-time social cue recognition system, a dialog/scenario manager, a behavior manager and a 3D rendering environment

II. LITERATURE REVIEW

Artificial Intelligence (AI) has recently become an important part of interview stimulation systems, drawing and growing attention from educators and professionals. These AI-based virtual interview simulators are effectively used to improve both communication and employability skills among

freshers and students who are seeking to get jobs. Several studies including those by Sharma et al. (2021) and Liu & Zhang (2022) shows that this system is designed into user interactive feedback that uses both verbal and nonverbal communication. This system observes speech patterns, body-language cues, and facial expressions captured in mock interview scenarios, our system also tracks the power of NLP (National Language Processing) to notice certain indicators of confidence, encompassing vocal tone, posture and eye contact. Through the careful interpretation of these multidimensional cues, our approach provides a clear understanding on how confident a person appears through an interview.

Our system can also generate dynamic, job specific interview questions based on a candidate's resume or a job description so that it can help the candidate furthermore to gain confidence. Unlike traditional mock interviews, which can be expensive these AI Stimulators provide instant, personalized feedback. They analyze response based on multiple factors including verbal, nonverbal communication and content.

The results obtained were encouraging and showed that the generated outcomes by the system can accurately detect the candidate's confidence. To be more precise about this point, Patel and Deshmukh (2023) reported that AI-based simulators not only improve the communication fluency but also increases the user's ability to explain technical concepts under pressure at the same time. Overall, AI-driven virtual interview simulators are gradually being used as beneficial in both academic and professional training, helping them to enhance their knowledge by strengthening their soft-skills as well as technical communication skills.

III. METHOD

The Virtual Interview Simulator using AI is designed as a web application that recreates a realistic interview environment for students and professionals. It consists aspects for question generation, speech processing, evaluation, and feedback. The system uses a react based interface for user interaction and a python backend that handles AI tasks such as speech recognition and natural language processing (NLP).

The question generation system produces specific and difficulty-based questions using a predefined dataset. These are generated using user's selected technical area. The user responses are then captured through microphone and converted to text using speech recognition applications such as Google Speech. Technical accuracy, clarity, and tone are evaluated by the NLP module processes through keyword detection, sentiment analysis etc. They are then marked by the evaluation engine, which combines content correctness and communication fluency to assess overall performance.

Finally, the feedback system generates reports representing the strengths and improvement areas. It also offers recommendations and sample models to help users improve their responses. The primary aim of the simulator is to improve confidence and communication quality through repeated practice sessions. The simulator improves gap between technical knowledge and effective communication.
for you.

Further, the research also points out that such simulators are capable of linking interview preparation as well as professional expectations. The current systems generally offer partial solutions but they do not actually cannot handle the real time difficulties. Also, some systems provide adaptive feedback that integrates based on

the previous history. This gap shows that there is a need for a single unified platform that combines these limitations to offer a broader and more realistic interview simulation experience.

Additionally, current studies point out that among technical skills communication— the ability to explain technical knowledge clearly is the most important cues for engineers and those who work with technology to pass interviews.

IV. RESULT AND DISCUSSIONS

The evaluation of the data gave an overall recognition rate of 88%. Some social cues like leaning back gave lower accuracies around 72%. The main reason for this slight dip in recognition outcomes are due to the unstable tracking methods of Microsoft Kinetic SDK and the camera's inability to perceive depth. Microsoft Kinetic is a low-cost option and hence lacks depth perception features. This capturing system is currently the best solution for the purpose. But the aim of the paper is not to provide a report on the recognition algorithms of these cameras but to use existing techniques to help prepare interviewees for interviews.

Most of the participants were positively surprised with the system and many felt that it was helpful for preparing for real world interviews. All participants found this method very effective as they could visualize when and what went wrong. All participants stated that they controlled their social and vocal cues in the start but as the interview proceeded their behavior quickly came back to their normal state. Some stated that it was not as real as a real interview but still better than a telephone interview.

Participants also pointed out some problems in the setup. The main issue was the timing delay between the answering of the interviewee and the

reaction of the interviewer. This hence, creates a disconnect at the side of the interviewee as the person feels as if the interviewer is impatient and sometimes interrupts the interviewee. The visual appearance of the interviewer was also criticized in some cases as it did not feel real enough. The reply time cannot be optimized as of now, until the neural network is optimized which will take larger sets of data and more time. The appearance of the interviewer can also not be changed to the extent to make it real as the amount of processing power to create a AI generated human looking interviewer is a lot and defeats the purpose of this study. But still some tweaks can be made in the model to make it look a more appealing and human like.

Some participants felt a bit uncomfortable as the table had to be kept too far away and they were not able to rest their hands making them feel unnatural. The rather large distance between the interviewee and the table was kept to ensure proper tracking with Microsoft Kinect. Solutions for this issue include different mounting points of these cameras or changing the type of camera or using multiple cameras (adding more cameras seems impractical as it drastically increases the computational power required and the cost of the project).

V. CONCLUSION & RECOMMENDATIONS

In this paper, we presented an effective web application to let the users improve their interviewing skills through a simulation of a real-time interview setting and generating situation- and user-specific questions. The model uses react as a user interface tool and an API that is utilized for the backend. It uses NLP (Natural Language Processing) to use for speech assessment. The user answers are recorded by audio devices using tools like Google Speech. Tone and clarity are

evaluated by the NLP model and parsed, tokenized and then passed through sentiment analysis. The questions are generated from a pre-defined set of data on an ordered scale of difficulty.

The responses are then evaluated with regard to correctness, fluency and other key parameters that would indicate the performance of the user in real interview situations. The feedback mechanism delivers the overall report including areas of strength and areas of weakness in the user's communication.

RECOMMENDATIONS:

Expand Communication Analysis: Simplify the simulator to analyze not only the content of technical response, but also the elements of delivery such as the quality of voice, pace, tone change, use of technical terms. The optional video input can enable the study of the body language and non-verbal expression.

Modify to Applied Interviews: Re-based Interviewing Modifications to suit other technical areas of work and job characteristics.

User validation: Perform pilot tests using facilitated interviews and collect the opinions of the users to confirm that the simulator is working. The actual experience on the ground would increase the feedback, the experience of the users and the rates of the interviews would offer support regarding the correct interview success rates.

Corrective Feedback: Have a system of providing feedback that will inform the users of what they are doing right and wrong in their technical communication.

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